

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Flanges

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R01	24.05.22	First Issue for Clients Review
A01	20.12.22	Issued for Approval

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- MPP-SPC-013

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Classes
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

Table of Contents

1.	INTRODUCTION	6
1.1.	Background Information	6
1.2.	Purpose	7
2.	REGULATIONS, CODES AND STANDARDS	8
3.	DESIGN	10
3.1.	Applicable Flange Type	10
3.2.	Wall Thickness	10
3.3.	Dimension	10
3.4.	Ends	10
3.5.	Bolt Hole	10
3.6.	Flange Facing Finish	11
3.7.	Repair and Rejection	11
4.	SPECIFIC MATERIAL REQUIREMENTS	12
4.1.	Carbon Steel and LTCS Flanges	12
4.2.	Substitution of equivalent materials	12
5.	INSPECTION & QUALITY CONTROL	13
5.1.	Material Certificates	13
5.2.	Certificate of Conformity	13
5.3.	Marking	13
5.4.	Packaging and Protection	13

Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
CS	Carbon Steel
ENP	Electroless Nickel Plated
ISO	International Organisation for Standardization
LTCS	Low Temperature Carbon Steel
MESC	Material & Equipment Standard and Code
Mmscf/d	Million standard cubic feet per day
MSS	Manufacturers Standardisation Society
MTO	Material Take Off
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
SPE	Society of Petroleum Engineers
SS	Stainless Steel

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

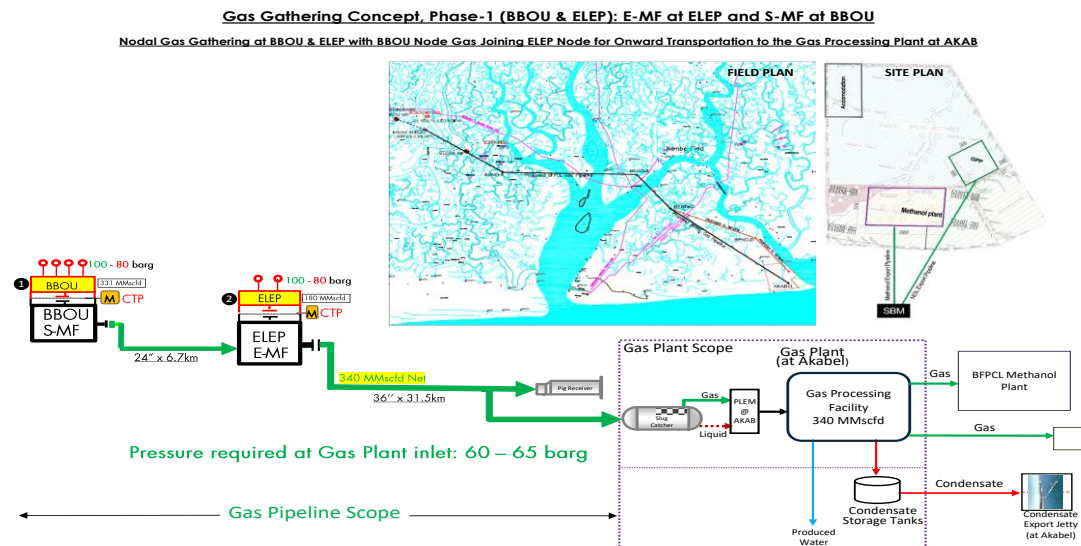


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification describes the technical requirements for the supply of flanges for BFPCL Phase 1 Project.

This specification applies to metallic flanges (i.e. flanges made of carbon steel and Stainless Steel,) and details key requirements for their design, fabrication, inspection and supply.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.25-Gen.	Piping – Process Design Requirements
Dep 31.38.01.10-Gen	Piping Classes – Basis Of Design
DEP 31.38.01.24-Gen.	Piping – Engineering And Layout Requirements
DEP 31.38.01.31-Gen	Shop And Field Fabrication Of Piping
DEP 31.38.01.84-Gen	Piping Classes
MESC SPE 76/210	Carbon Steel Forgings (Amendments/Supplements To ASTM A 105)
MESC SPE 76/101	Flanges (Amendments/Supplements To ASME B16.47)
MESC SPE 76/212	Alloy And Stainless Steel Forgings (Amendments/Supplements To ASTM A 182)
MESC SPE 76/100	Flanges (Amendments/Supplements To ASME B16.5)
ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B16.25	Butt Welding Ends
ASME B16.47	Large Diameter Steel Flanges: NPS 26 through NPS 60 Metric/Inch Standard.
ASME B31.3	Process Piping
ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
MSS SP-97	Integrally Reinforced Forged Branch Outlet Fittings-Socket Welding, Threaded, and Butt-Welded Ends
MSS SP43	Wrought stainless steel butt-welding fittings. Wrought and Fabricated Butt-Welding Fittings for Low Pressure, Corrosion
ASTM A182	Standard Specification for Forged or Rolled Alloy and Stainless-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A105	Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A234	Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service.

The latest versions of the regulations, codes, and standards and extant amendments or supplements

shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Design

Flanges shall be design in accordance with ASME B16.5 for diameter ½" up to 24" and ASME B16.47 series A for diameters of 26" and above.

3.1. Applicable Flange Type

The following flange types shall be used in the construction of piping system for the project.

For NPS 2 and above, flange type shall be weld neck with a raised-face (RF) for class 600# and below while class 900# and above shall be Ring Type Joint (RTJ).

For NPS 3/4 to NPS 1-1/2, the raised face flanges shall have socket welding ends for all pressure classes.

3.2. Wall Thickness

The hub thickness of the flange shall be the same with the adjoining pipes and as specified in the material requisitions. Where it becomes imperative for Vendors to procure a different wall thickness for a particular flange due to manufacturing limitations, the proposed wall thickness shall be of standard as per ASME B36.10M and must not be smaller than the wall thickness specified on the material requisition.

In their Quotation the Bidder is required to clearly state whether the nominated thicknesses can be supplied or alternatively, specify the next thicknesses they are able to supply.

3.3. Dimension

The end thicknesses and the bore dimensions of weld-neck flanges shall match the adjoining pipe schedule in accordance with ASME B36.10M.

End to end dimensions of weld-neck flanges shall comply with ASME B16.5 for diameters up to 24 inches. The dimensions of weld-neck flanges and blind flanges 26 inches or more in diameter shall be in accordance with ASME B16.47 Series A.

3.4. Ends

Weld-neck flanges up to 24" in diameter shall have bevelled ends prepared according to the requirements ASME B16.5.

The ends of flanges 26" or more in diameter shall comply with ASME B16.47 Series A. Series B shall not be used in this project.

The ends of socket-welding flanges shall comply with ASME B16.5.

3.5. Bolt Hole

Bolt holes shall be in multiples of four. Bolt holes shall be equally spaced. Drilling Bolt holes shall

be made by multi-spindle drills. The spot facing of the flanges at the bolt holes shall be in accordance with MSS SP 9.

3.6. Flange Facing Finish

The finish of the gasket contact faces shall be in accordance with ASME B16.5 or ASME B16.47, as applicable. For Raised face Flanges with Spiral wound gaskets, a serrated (spiral or concentric) finish with an average roughness value between 3.2 to 6.3 μm shall be furnished and judged by visual comparison with Ra standards according to ASME B46.1, not by instruments having stylus tracers and electronic amplification.

For RTJ Flanges, the groove shall be very smooth with an average roughness of 1.6 μm

3.7. Repair and Rejection

Repairs of defects and face imperfections shall only be permitted if performed fully in accordance with the reference codes and standards and with the approval of the purchaser.

Imperfections in the flange facing finish shall not exceed the dimensions shown in Table 3 (Table F3 of Annex F) of ASME B16.5 otherwise flanges not meeting the design specification shall be rejected.

4. Specific Material Requirements

4.1. Carbon Steel and LTCS Flanges

All flanges shall be hot forged and shall undergo an appropriate heat treatment after forging in accordance with the requirements of their reference codes and standards.

By product analysis (ladle Analysis), carbon equivalent shall be less than 0.43 calculated using the formula:

$$\text{C.E.} = \text{C} + \frac{\text{Mn}}{6} + \frac{\text{Cr} + \text{Mo} + \text{V}}{5} + \frac{\text{Ni} + \text{Cu}}{15}$$

Carbon steels, including normalized steels shall have a maximum carbon content of 0.23%.

If the content of alloying elements other than C or Mn is unknown, the formula to be used shall be:

$$\text{C.E.} = \text{C} + \frac{\text{Mn}}{6}$$

In this case, the carbon equivalent shall not be more than 0.40.

All carbon steel materials must be impact tested at minus 29°C (-20°F) in accordance with ASME B31.3 (paragraph 323.3), however, carbon steels may not be tested if their nominal neck wall thickness is not above 12.7 mm (½ inch). Otherwise, Impact-tested steel SHALL be used for carbon steel flanges which lower design temperature is below 0 °C (32 °F).

All carbon steel materials for services at lower temperatures (i.e., below minus 29°C) shall undergo toughness testing in accordance with the requirements of their reference codes and standards. Flange material specification for low temperature service shall be ASTM A350 Grade LF2 class 1

Austenitic stainless-steel flanges Type 321 stainless steel shall have a carbon content greater than or equal to 0.02%. Austenitic stainless-steel flanges shall undergo an annealing heat treatment in accordance with the corresponding ASTM standards.

where galvanised flanges are specified, it shall be Hot Deep Galvanized. Electro-deposited coating of zinc is not acceptable.

4.2. Substitution of equivalent materials

Substitution of flanges of a specified schedule by a higher schedule must be submitted to the EMPLOYER for approval. In this case, internal tapering will be necessary to match the specified schedule. The slope of this tapering shall comply with the reference standards.

5. Inspection & Quality control

The inspection and tests necessary to confirm that the products meet the requirements of the standards, specifications and requisitions shall be carried out in the Manufacturer's plant by qualified personnel who shall be independent of the production department of the plant.

The Inspector shall check for the following conformities;

- Shapes and nominal dimensions
- Dimensional tolerances
- Surface treatment and finish conditions
- Marking and finishing conditions.

5.1. Material Certificates

Manufacturer shall submit EN 10204 3.1 material certificates giving the results of the chemical analysis and of the mechanical tests carried out in accordance with the requirements of the reference codes and standards.

5.2. Certificate of Conformity

Vendor shall submit certificates of conformity with ASTM standards specified in the requisition for all flanges.

5.3. Marking

The following information shall be marked by cold stamping on each flange as per MSS SP 25;

- The specification and grade of material
- The heat number
- The batch number
- The rating
- The diameter

5.4. Packaging and Protection

The bevelled ends and contact faces of the flanges shall be protected by a peel-off varnish.

All stainless-steel flanges shall be protected against the risk of saline corrosion during cleaning, transport, fabrication and storage.

The flanges must be packed in sea packing, which must, in addition, be water-proof in the case of austenitic stainless steels.

Packing must be suitable for all stages of transport.